FetchFood

DATA 5300 COURSE PROJECT FOR A/B TESTING

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BACKGROUND AND PURPOSE

- **Company Overview:**
- FetchFood is a Seattle-based start-up connecting restaurants and grocery stores with customers for food deliveries.

- **Unique Business Model:**
- Unlike competitors like UberEats, FetchFood allows drivers to cancel their deliveries at any time without penalties.
- This driver flexibility impacts the overall efficiency and financial health of FetchFood and its partner businesses.

- **Challenge Identified:**
- High rates of delivery cancellations lead to operational inefficiencies and financial strain on FetchFood and the partnered restaurants/grocery stores.

- **Objective:**
- Assess the effectiveness of imposing cancellation fees on reducing delivery cancellations.
- Identify the optimal penalty level that minimizes cancellations without adversely affecting driver participation or customer satisfaction.

- **Experiment Design**:
- Duration: Six weeks.

- **Groups Tested:**
- Control Group: No cancellation fee ($0).
- Treatment Group 1: $10 cancellation fee.
- Treatment Group 2: $20 cancellation fee.

- **Expected Outcome:**
- A reduction in the number of cancellations with the introduction of cancellation fees.

METRICS AND METHODS

Short-Term Metrics:

- **Canceled Orders per Driver:**
  - Tracks the average number of orders canceled by each driver across different penalty variant.

- **Accepted Orders per Driver:**
  - Measures the average number of completed orders per driver across different penalty variant.

- **Revenue per Driver:**
  - Evaluates the revenue generated by each driver based on completed orders during the same period.

Methods used for analysis:

- **ANOVA (Analysis of Variance):** Compares the means of metrics across multiple groups to detect significant differences.
- **Welch's T-Test with Bonferroni Correction:** Performs pairwise comparisons while adjusting for unequal variances and multiple testing.
- **Confidence Intervals:** Estimates the range within which a metric's true mean likely falls, providing precision insights.
- **Linear Regression:** Models the relationship between penalty levels and metrics to quantify impact and trends.
- **Z-Score Normalization:** Standardizes metric values to measure deviations from the mean for normalized comparisons.

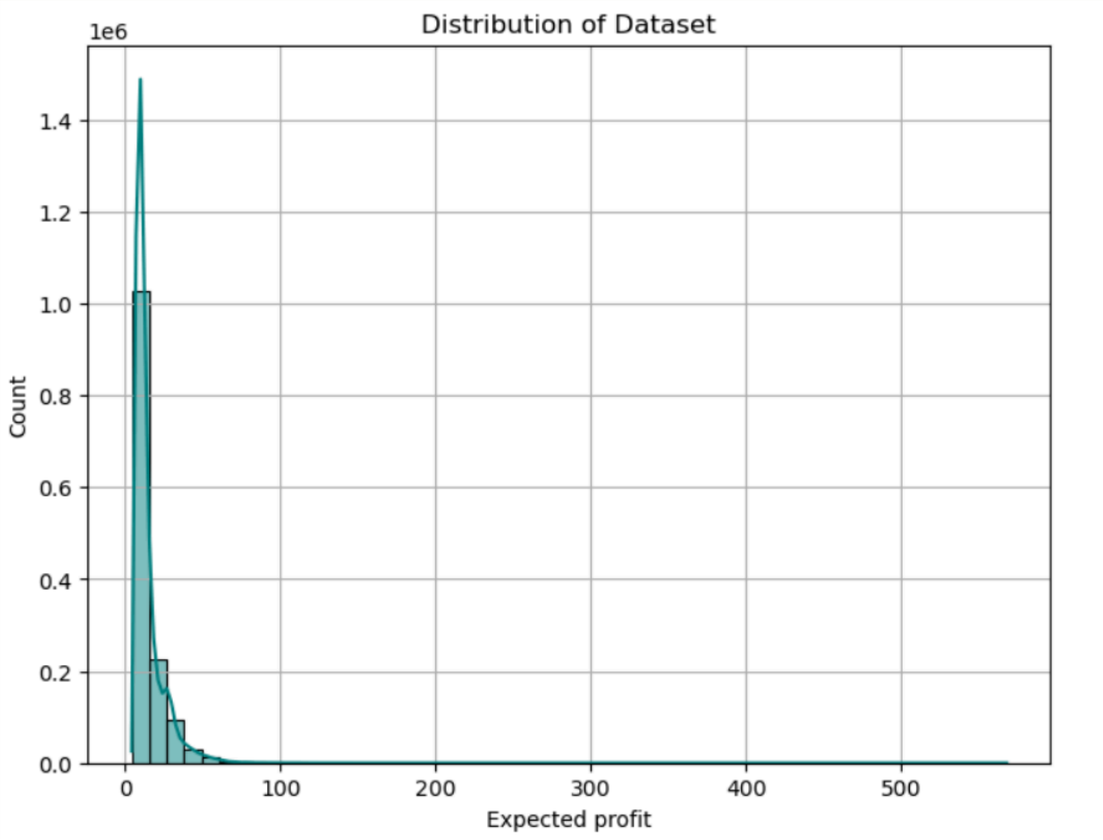
Long-Term Metrics:

- **Driver Retention:**
  - The rate at which drivers continue to work for FetchFood over time.

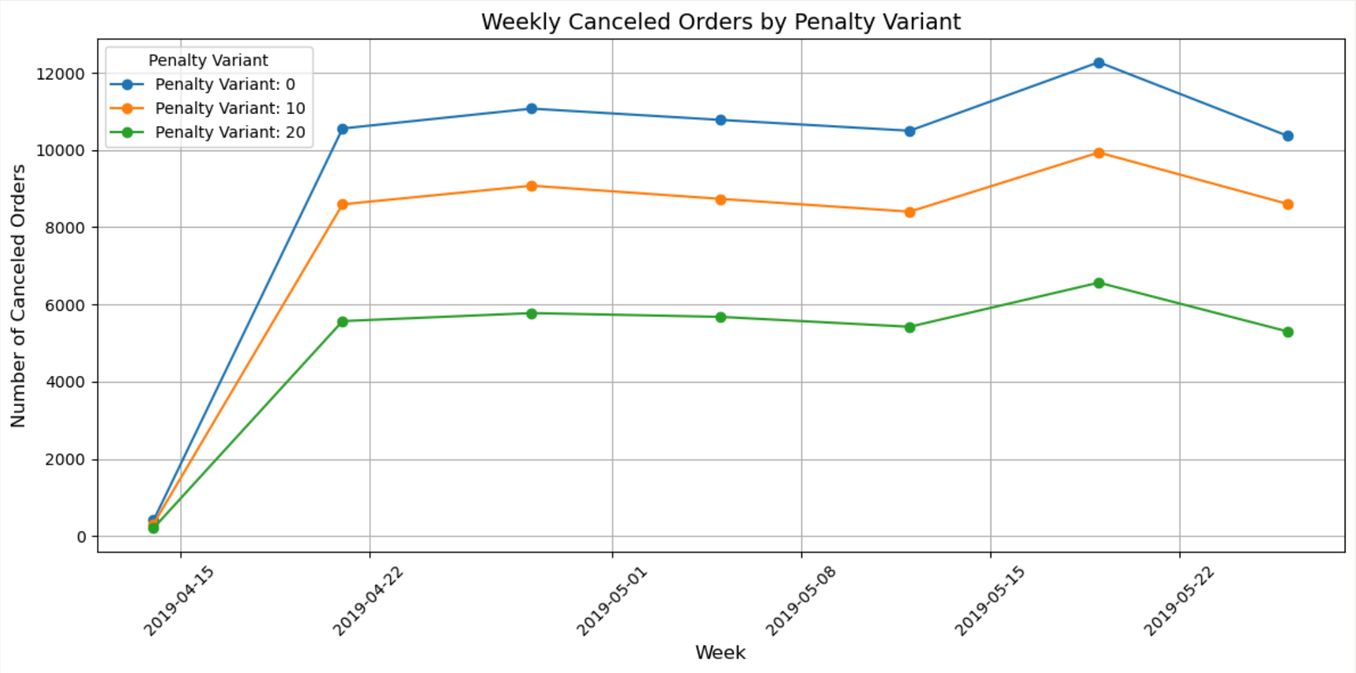
- **Operational Efficiency:**
  - The effectiveness of processes in delivering food with minimal delays and cost.

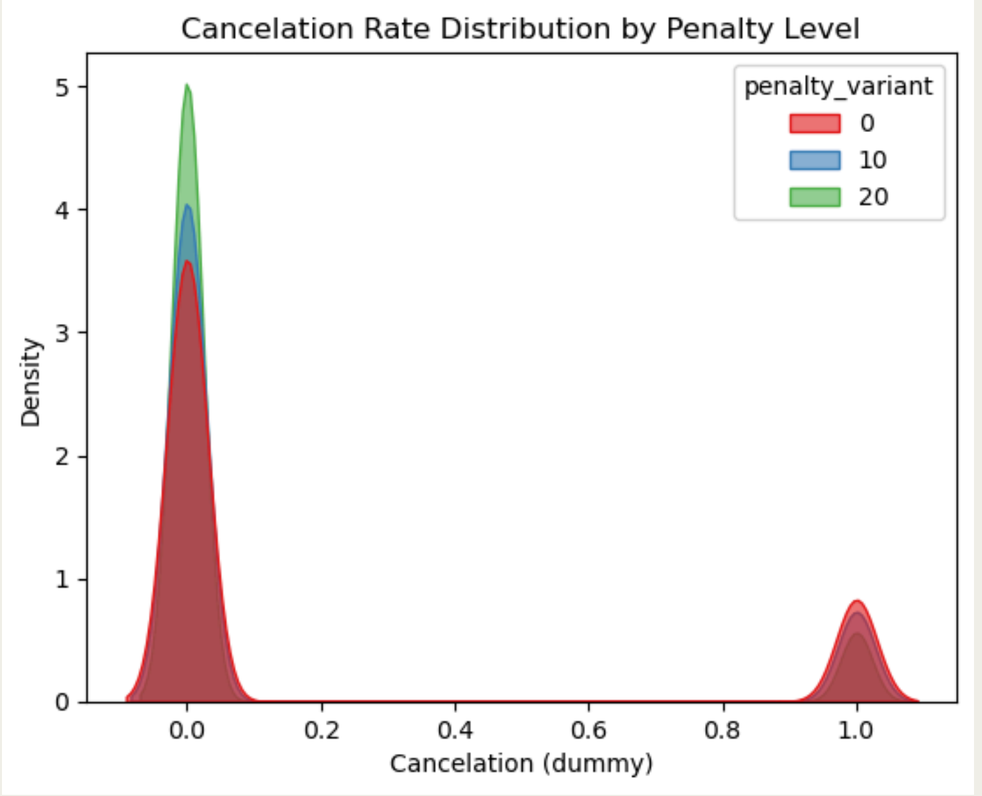
Note:
The dataset covered six weeks of time duration, which was inadequate to perform long term analysis.

ANALYSIS - DATA VISUALIZATION






Weekly Canceled Orders by Penalty Variant
Penalty Variant
Penalty Variant: 0
Penalty Variant: 10
Penalty Variant: 20
Number of Canceled Orders
0
2000
4000
6000
8000
10000
12000
2019-04-15
2019-04-22
2019-05-01
2019-05-08
2019-05-15
2019-05-22
Week


Cancelation Rate Distribution by Penalty Level
penalty_variant
0
10
20
Density
0
1
2
3
4
5
0.0
0.2
0.4
0.6
0.8
1.0
Cancelation (dummy)

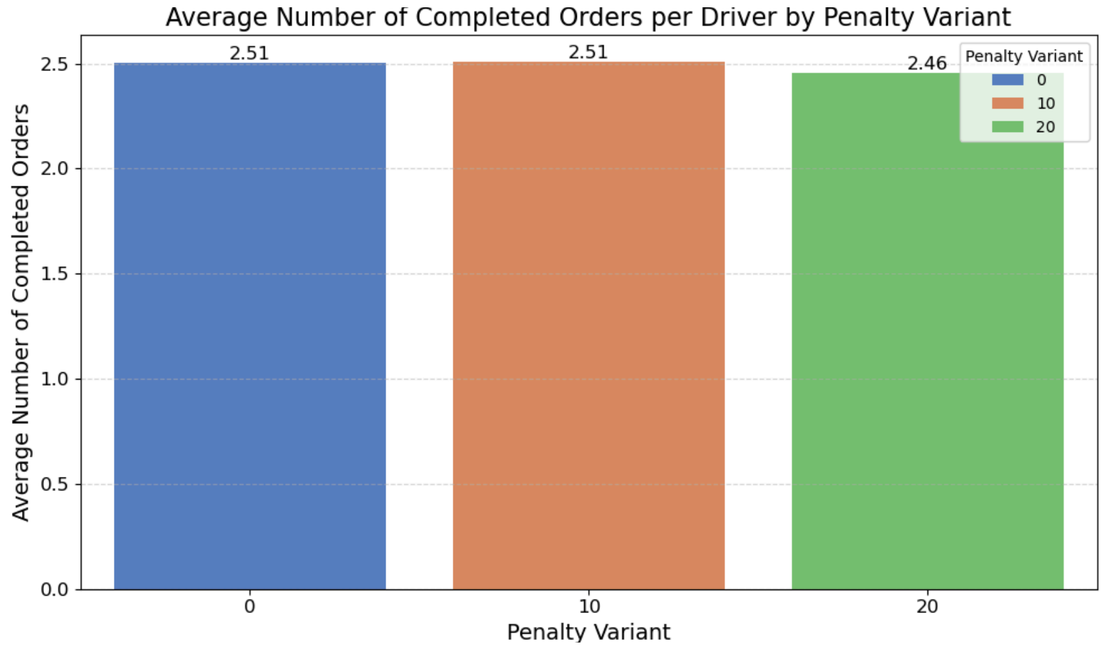
ANALYSIS - A/B TESTING

ANOVA Test Results:

Hypothesis	Test	P-Value	Conclusion
H_0: No difference in mean completed orders. H_a: At least one variant differs.	Completed Orders	1.1186e-06	Significant difference in mean completed orders.
H_0: No difference in mean canceled orders. H_a: At least one variant differs.	Canceled Orders	1.1186e-06	Significant difference in mean canceled orders.
H_0: No difference in mean revenue. H_a: At least one variant differs.	Revenue per Driver	9.3623e-291	Significant difference in mean revenue.

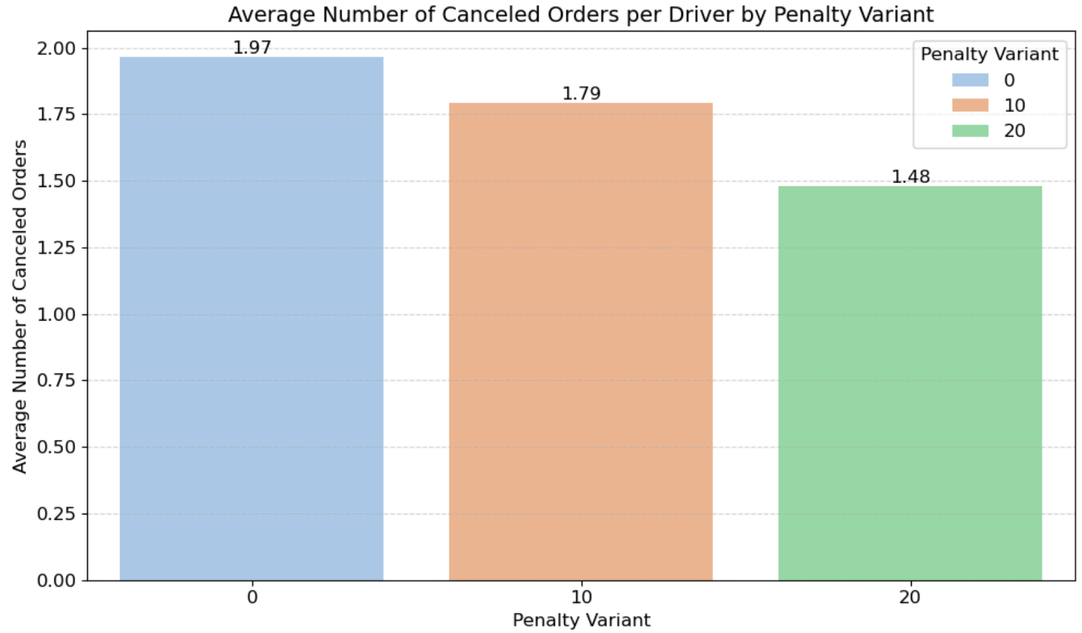
Multiple t-tests with Bonferroni Adjustments for **Completed Orders:**

Hypothesis	Comparison	T-Statistics	P-Value	Bonferroni Adjusted P-Value	Conclusion
H_0: No difference in mean completed orders. H_a: Difference exists.	(0 vs. 10)	-0.9692	0.3325	0.9974	No significant difference.
H_0: No difference in mean completed orders. H_a: Difference exists.	(0 vs. 20)	4.4446	8.81e-06	2.6431e-05	Significant difference.
H_0: No difference in mean completed orders. H_a: Difference exists.	(10 vs. 20)	5.5065	3.6625e-08	1.0988e-07	Significant difference.



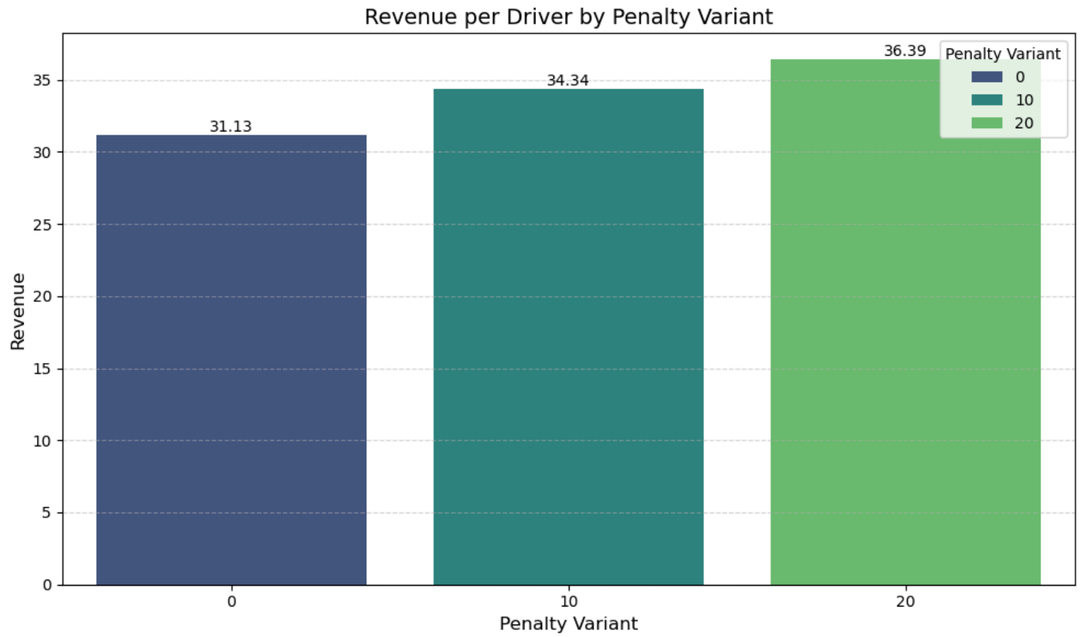
Multiple t-tests with Bonferroni Adjustments for **Canceled Orders**:

Hypothesis	Comparison	T-Statistics	P-Value	Bonferroni Adjusted P-Value	Conclusion
H_0: No difference in mean canceled orders. H_a: Difference exists.	(0 vs. 10)	9.9679	2.1919e-23	6.5758e-23	Significant difference.
H_0: No difference in mean canceled orders. H_a: Difference exists.	(0 vs. 20)	32.4899	2.8851e-229	8.6552e-229	Significant difference.
H_0: No difference in mean canceled orders. H_a: Difference exists.	(10 vs. 20)	22.5548	4.4175e-112	1.3253e-111	Significant difference.



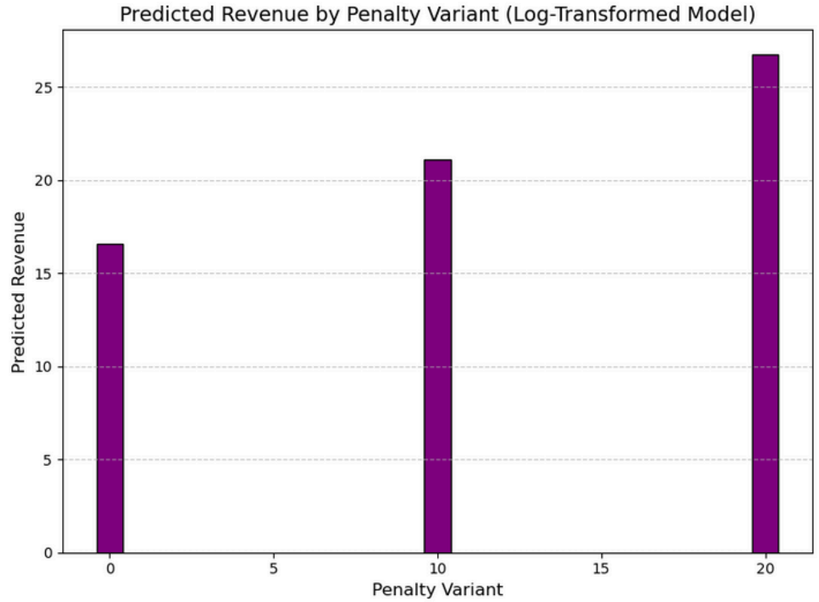
Multiple t-tests with Bonferroni Adjustments for **Revenue**:

Hypothesis	Comparison	T-Statistics	P-Value	Bonferroni Adjusted P-Value	Conclusion
H_0: No difference in mean revenue. H_a: Difference exists.	(0 vs. 10)	-22.1192	2.5892e-108	7.7676e-108	Significant difference.
H_0: No difference in mean revenue. H_a: Difference exists.	(0 vs. 20)	-36.3940	2.7647e-289	8.2942e-289	Significant difference.
H_0: No difference in mean revenue. H_a: Difference exists.	(10 vs. 20)	-14.1243	2.7956e-45	8.3867e-45	Significant difference.



Linear Regression Results:

Hypothesis	Variable	Coefficient	Interpretation
H_0: No effect of penalty variant on cancellations. H_a: Penalty variant affects cancellations.	Penalty Variant	-0.0050	Higher penalties reduce canceled orders.
H_0: No effect of penalty variant on revenue. H_a: Penalty variant affects revenue.	Penalty Variant	0.0228	Higher penalties increase revenue.

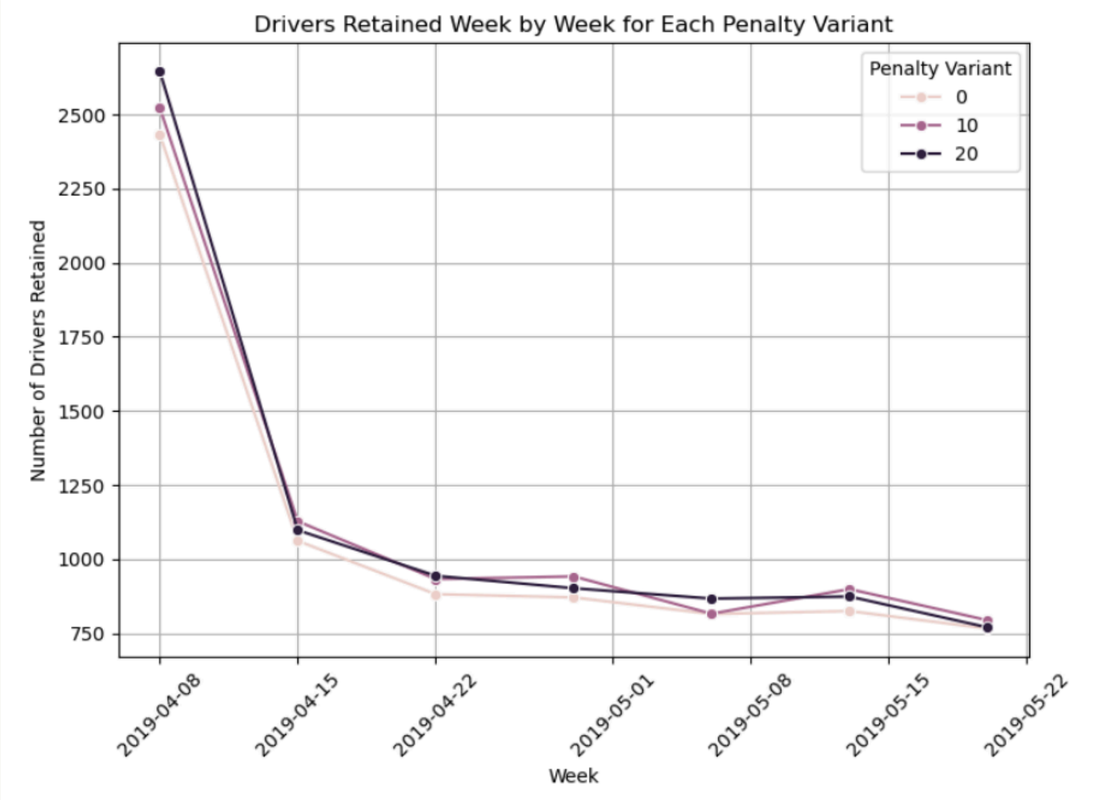


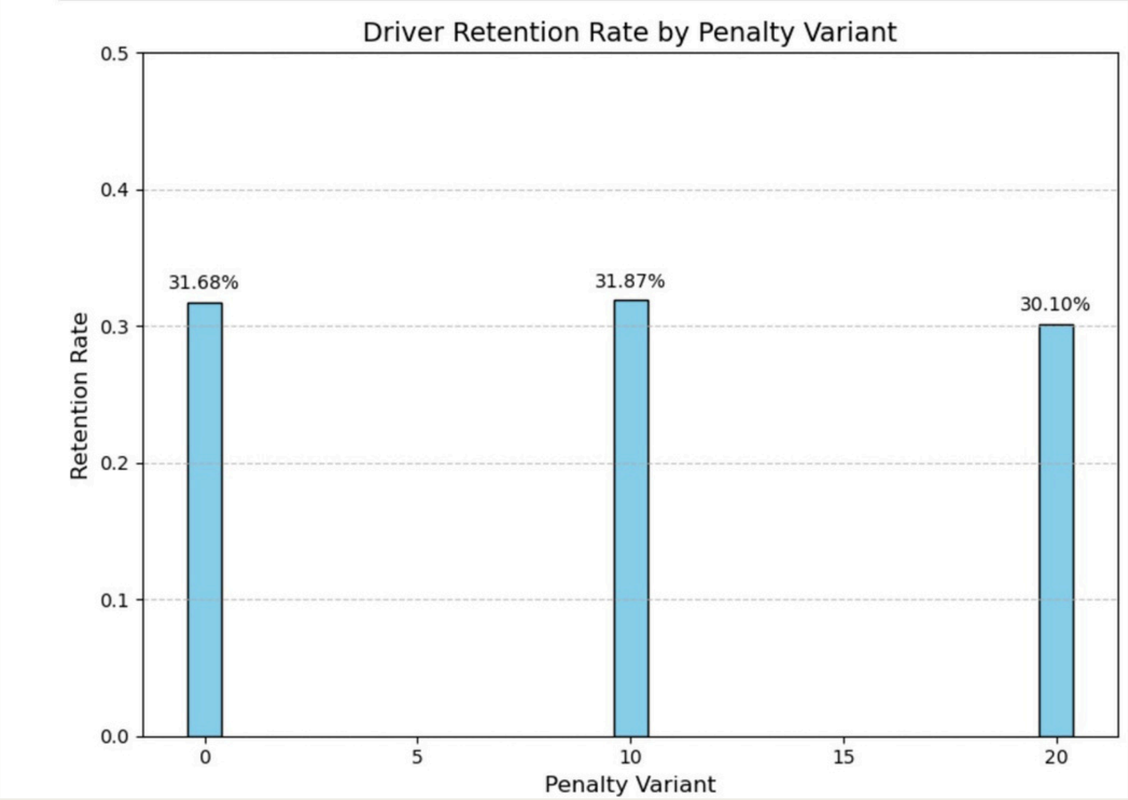


Z-Score Normalization Results:

Hypothesis	Z-Score (Revenue)	Z-Score (Canceled Orders)	Composite Score	Penalty Variant	Interpretation
H_0: No optimal penalty variant. H_a: One variant is optimal.	-1.0679	0.8828	-0.9754	20	Optimal penalty balancing revenue and minimizing cancellations.
H_0: No optimal penalty variant. H_a: One variant is optimal.	0.1535	0.2031	-0.0248	10	Moderate performance in revenue and cancellations.
H_0: No optimal penalty variant. H_a: One variant is optimal.	0.9144	-1.0860	1.0002	0	Least effective in balancing revenue and cancellations.

ANALYSIS - LONG TERM





Above plots show that $20 penalty leads to driver dissatisfaction in the longer run.

CONCLUSIONS AND RECOMMENDATIONS

Statistical Results:

- $20 penalty minimizes cancellations.
- $20 penalty maximizes short-term revenue.

Driver Engagement:

- $20 penalty leads to lower driver retention.
- Graphs show increased dissatisfaction among drivers.

Conclusion:

- $20 is best for short-term results but unsustainable long-term.
- $20 shows driver dissatisfaction in long-term.

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Recommendations:

- Instead of imposing a fixed penalty, we can implement a system that monitors each driver's monthly cancellation count and calculates a proportional penalty based on their individual cancellation rate. This approach ensures fairness and accountability while encouraging drivers to minimize unnecessary cancellations.

- Drivers with fewer cancellations can be incentivized through rewards or coupons, recognizing their reliability and encouraging consistent performance.

- After imposing the penalty system, take regular feedback from drivers as well as customers.

SUGGESTIONS FOR FUTURE EXPERIMENTS

- Expand treatment groups to more penalty variants.
- Historical data can be provided to A/B Test metrics before and after penalty was enforced (CUPED).
- Data can differentiate between part-time and full-time drivers.
- Dataset can also include weather and geography as a metric.
- We can include surge hours in our analysis if locations are provided.
- We can also include driver ratings by customers as a metric to calculate driver-rider engagement and customer satisfaction.

Thank you!

